

DTSA 5741

Modeling Climate Anomalies with Statistical Analysis

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	Machine Learning Engineer (70.6 / 100)	46.8 / 100
SPECIALIZATION Climate Change: Time Series Analysis		

47	79	+41
CURRENT AVG SCORE across 6 job roles	POTENTIAL SCORE with recommended additions	MAX SCORE GAIN best single-role improvement

CURRENT COURSE DESCRIPTION

This course introduces the use of statistical analysis in Python programming to study and model climate data, specifically with the SciPy and NumPy package. Topics include data visualization, predictive model development, simple linear regression, multivariate linear regression, multivariate linear regression with interaction, and logistic regression. Strong emphasis will be placed on gathering and analyzing climate data with the Python programming language.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	70.6	96.8	+26.1	MEDIUM
ML Engineer	50.1	78.1	+28.0	MEDIUM
Data Scientist	47.5	85.4	+37.8	HIGH
AI Engineer	46.4	76.7	+30.3	HIGH
Applied Scientist	36.6	77.2	+40.7	HIGH
MLOps Engineer	29.4	60.5	+31.1	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

Deep Learning

1 of 6 roles

training

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

LOW

Optional: Deep Learning — would broaden appeal to one additional role. Candidate terms: training.

GAP KEYWORDS BY ROLE

Machine Learning Engineer	70.6 → 96.8 (+26.1 pts)
sets learning tools exploratory design science models applications	
ML Engineer	50.1 → 78.1 (+28.0 pts)
learning machine models technical global history terms tools	
Data Scientist	47.5 → 85.4 (+37.8 pts)
toolkit sets different behaviors social	
AI Engineer	46.4 → 76.7 (+30.3 pts)
models learning technical machine design tools policy terms	
Applied Scientist	36.6 → 77.2 (+40.7 pts)
biology computational research literature models technical energy learning	
MLOps Engineer	29.4 → 60.5 (+31.1 pts)
learning machine training tools models	

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum–Industry Alignment project team.